Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-6)

BIOPHARMACEUTICS AND PHARMACOKINETICS (THEORY)

Subject Code : BP604T

M.Code : 77989

Date of Examination: 20-05-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.

SECTION-A

1. **Explain briefly:**

   a) What is the difference between one and two compartment models?

   b) Draw a typical plasma concentration Vs time plot for oral drug administration and indicate the pharmacokinetic parameters.

   c) What is the difference between elimination and excretion rate constants?

   d) What are the most bioavailable dosage forms orally and parenterally?

   e) Name an acidic and basic plasma protein.

   f) Give two examples of therapeutic equivalents.

   g) What is meant by generic substitutes? Give two examples.

   h) What is meant by pharmaceutical alternative? Give two examples.

   i) Enumerate the advantages of IV infusion administration. .

   j) How is AUC calculated?

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SECTION-B

2. Describe the plasma pharmacokinetics of a drug following one-compartment open model after oral administration by deriving suitable equations. How is Ka and Ke calculated from these derivations.

3. Discuss the physiological factors influencing drug absorption.

4. What is meant by non-linear pharmacokinetics? Discuss the various reasons for non-linear behaviour of drugs.

SECTION-C

5. Distinguish between absolute and relative bioavailability.

6. Distinguish between facilitated and passive drug transport.

7. Distinguish between phase-I and phase-II metabolism.

8. Briefly discuss the factors affecting renal drug clearance.

9. Mention the clinical implications of protein binding on drug pharmacokinetics.

10. Mention the units of Ka, Clearance, Vd, Cmax and AUC.

11. Give examples of five drugs whose bioavailability is influenced due to limited solubility.

12. Mention the factors influencing gastric emptying time.

13. Discuss IVIVC models.

NOTE : Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UMC against the Student.

2025